

Creation Date 19-Apr-2018

Revision Date 23-May-2024

Revision Number 4

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: Triisobutylaluminum, 25% w/w in hexane  
Cat No. : **89057**  
Molecular Formula C<sub>12</sub> H<sub>27</sub> Al

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

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|                                                                        |                     |
|------------------------------------------------------------------------|---------------------|
| Flammable liquids                                                      | Category 2 (H225)   |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1 (H260)   |
| Pyrophoric liquids                                                     | Category 1 (H250)   |
| <b><u>Health hazards</u></b>                                           |                     |
| Aspiration Toxicity                                                    | Category 1 (H304)   |
| Skin Corrosion/Irritation                                              | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation                                      | Category 1 (H318)   |
| Reproductive Toxicity                                                  | Category 2 (H361f)  |
| Specific target organ toxicity - (single exposure)                     | Category 3 (H336)   |
| Specific target organ toxicity - (repeated exposure)                   | Category 2 (H373)   |
| <b><u>Environmental hazards</u></b>                                    |                     |
| Chronic aquatic toxicity                                               | Category 2 (H411)   |

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### Hazard Statements

H225 - Highly flammable liquid and vapor  
H250 - Catches fire spontaneously if exposed to air  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H304 - May be fatal if swallowed and enters airways  
H314 - Causes severe skin burns and eye damage  
H336 - May cause drowsiness or dizziness  
H361f - Suspected of damaging fertility  
H373 - May cause damage to organs through prolonged or repeated exposure  
H411 - Toxic to aquatic life with long lasting effects  
EUH014 - Reacts violently with water

### Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

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## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component                       | CAS No   | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                        |
|---------------------------------|----------|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hexane                          | 110-54-3 | EEC No. 203-777-6 | 75.00    | Flam. Liq. 2 (H225)<br>Asp. Tox. 1 (H304)<br>Skin Irrit. 2 (H315)<br>STOT SE 3 (H336)<br>Repr. 2 (H361f)<br>STOT RE 2 (H373)<br>Aquatic Chronic 2 (H411) |
| Aluminum, tris(2-methylpropyl)- | 100-99-2 | EEC No. 202-906-3 | 25.00    | Pyr. Liq. 1 (H250)<br>Water-react. 1 (H260)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>(EUH014)                                                     |

| Component | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------|---------------------------------------|----------|-----------------|
| Hexane    | STOT RE 2 (H373) :: C>=5%             | -        | -               |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                   |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                   |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                             |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. Risk of serious damage to the lungs (by aspiration). |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                    |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated:

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Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## 4.3. Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Dry sand. Carbon dioxide (CO<sub>2</sub>). Powder. Do not use water or foam. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

Water.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Metal oxides, Isobutane.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

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Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 7.2. Conditions for safe storage, including any incompatibilities

Corrosives area. Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Storage Class/LGK 4.2

Switzerland - Storage of hazardous substances

Storage class - SC 4.2  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component                       | European Union                                       | The United Kingdom                                                                      | France                                                                                                                                                                                                     | Belgium                                                | Spain                                                                          |
|---------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------|
| Hexane                          | TWA: 20 ppm (8hr)<br>TWA: 72 mg/m <sup>3</sup> (8hr) | TWA: 72 mg/m <sup>3</sup><br>TWA: 20 ppm<br>STEL: 60 ppm<br>STEL: 216 mg/m <sup>3</sup> | TWA / VME: 20 ppm (8 heures). restrictive limit<br>TWA / VME: 72 mg/m <sup>3</sup> (8 heures). restrictive limit<br>TWA / VME: 1000 mg/m <sup>3</sup> (8 heures).<br>STEL / VLCT: 1500 mg/m <sup>3</sup> . | TWA: 20 ppm 8 uren<br>TWA: 72 mg/m <sup>3</sup> 8 uren | TWA / VLA-ED: 20 ppm (8 horas)<br>TWA / VLA-ED: 72 mg/m <sup>3</sup> (8 horas) |
| Aluminum, tris(2-methylpropyl)- |                                                      | STEL: 6 mg/m <sup>3</sup> 15 min<br>TWA: 2 mg/m <sup>3</sup> 8 hr                       | TWA / VME: 2 mg/m <sup>3</sup> (8 heures).                                                                                                                                                                 |                                                        |                                                                                |

| Component | Italy                                                                                                    | Germany                                   | Portugal                                                         | The Netherlands                                                            | Finland                                                               |
|-----------|----------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Hexane    | TWA: 20 ppm 8 ore.<br>Time Weighted Average<br>TWA: 72 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average | TWA: 180 mg/m <sup>3</sup><br>TWA: 50 ppm | TWA: 20 ppm 8 horas<br>TWA: 72 mg/m <sup>3</sup> 8 horas<br>Pele | STEL: 144 mg/m <sup>3</sup> 15 minuten<br>TWA: 72 mg/m <sup>3</sup> 8 uren | TWA: 20 ppm 8 tunteina<br>TWA: 72 mg/m <sup>3</sup> 8 tunteina<br>Iho |

| Component | Austria             | Denmark             | Switzerland | Poland                      | Norway              |
|-----------|---------------------|---------------------|-------------|-----------------------------|---------------------|
| Hexane    | MAK-KZGW: 80 ppm 15 | TWA: 20 ppm 8 timer | Haut/Peau   | TWA: 72 mg/m <sup>3</sup> 8 | TWA: 20 ppm 8 timer |

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|                                    |                                                                                                                                        |                                                                                                                |                                                                                                                                                  |           |                                                                                                                                                                                                                       |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | Minuten<br>MAK-KZGW: 288 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 20 ppm 8<br>Stunden<br>MAK-TMW: 72 mg/m <sup>3</sup> 8<br>Stunden | TWA: 72 mg/m <sup>3</sup> 8 timer<br>STEL: 40 ppm 15<br>minutter<br>STEL: 144 mg/m <sup>3</sup> 15<br>minutter | STEL: 400 ppm 15<br>Minuten<br>STEL: 1440 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 50 ppm 8<br>Stunden<br>TWA: 180 mg/m <sup>3</sup> 8<br>Stunden | godzinach | TWA: 72 mg/m <sup>3</sup> 8 timer<br>TWA: 40 ppm 8 timer<br>TWA: 275 mg/m <sup>3</sup> 8 timer<br>STEL: 30 ppm 15<br>minutter. value<br>calculated<br>STEL: 108 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated |
| Aluminum,<br>tris(2-methylpropyl)- |                                                                                                                                        |                                                                                                                | TWA: 2 mg/m <sup>3</sup> 8<br>Stunden                                                                                                            |           | TWA: 2 mg/m <sup>3</sup> 8 timer                                                                                                                                                                                      |

| Component | Bulgaria                                   | Croatia                                                                            | Ireland                                                                                                                      | Cyprus                                   | Czech Republic                                                                                                      |
|-----------|--------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Hexane    | TWA: 20 ppm<br>TWA: 72.0 mg/m <sup>3</sup> | kože<br>TWA-GVI: 20 ppm 8<br>satima.<br>TWA-GVI: 72 mg/m <sup>3</sup> 8<br>satima. | TWA: 20 ppm 8 hr.<br>TWA: 72 mg/m <sup>3</sup> 8 hr.<br>STEL: 60 ppm 15 min<br>STEL: 216 mg/m <sup>3</sup> 15<br>min<br>Skin | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 70 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 200 mg/m <sup>3</sup> |

| Component | Estonia                                                                | Gibraltar                                          | Greece                                   | Hungary                                                                                 | Iceland                                                                                                                               |
|-----------|------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Hexane    | TWA: 20 ppm 8<br>tundides.<br>TWA: 72 mg/m <sup>3</sup> 8<br>tundides. | TWA: 20 ppm 8 hr<br>TWA: 72 mg/m <sup>3</sup> 8 hr | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 72 mg/m <sup>3</sup> 8<br>órában. AK<br>lehetséges borön<br>keresztüli felszívódás | TWA: 20 ppm 8<br>klukkustundum.<br>TWA: 72 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Ceiling: 40 ppm<br>Ceiling: 144 mg/m <sup>3</sup> |

| Component | Latvia                                   | Lithuania                                          | Luxembourg                                                         | Malta                                    | Romania                                              |
|-----------|------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------|------------------------------------------|------------------------------------------------------|
| Hexane    | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 20 ppm IPRD<br>TWA: 72 mg/m <sup>3</sup> IPRD | TWA: 20 ppm 8<br>Stunden<br>TWA: 72 mg/m <sup>3</sup> 8<br>Stunden | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 20 ppm 8 ore<br>TWA: 72 mg/m <sup>3</sup> 8 ore |

| Component | Russia                                                        | Slovak Republic                                                                          | Slovenia                                                                                                                           | Sweden                                                                                                                                                                | Turkey                                                 |
|-----------|---------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Hexane    | TWA: 300 mg/m <sup>3</sup> 0780<br>MAC: 900 mg/m <sup>3</sup> | Ceiling: 140 mg/m <sup>3</sup><br>TWA: 20 mg/m <sup>3</sup><br>TWA: 72 mg/m <sup>3</sup> | TWA: 20 ppm 8 urah<br>TWA: 72 mg/m <sup>3</sup> 8 urah<br>STEL: 576 mg/m <sup>3</sup> 15<br>minutah<br>STEL: 160 ppm 15<br>minutah | Binding STEL: 50 ppm<br>15 minuter<br>Binding STEL: 180<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 20 ppm 8 timmar.<br>NGV<br>TLV: 72 mg/m <sup>3</sup> 8<br>timmar. NGV | TWA: 20 ppm 8 saat<br>TWA: 72 mg/m <sup>3</sup> 8 saat |

## Biological limit values

List source(s):

| Component | European Union | United Kingdom | France                                                      | Spain                                                 | Germany                                                                                                |
|-----------|----------------|----------------|-------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Hexane    |                |                | 2,5-Hexanedione: 5<br>mg/g creatinine urine<br>end of shift | 2,5-Hexanedione: 0.2<br>mg/L urine end of<br>workweek | 2,5-Hexandione plus<br>4,5-Dihydroxy-2-hexano<br>ne (after hydrolysis): 5<br>mg/L urine (end of shift) |

| Component | Italy | Finland | Denmark | Bulgaria | Romania                                                   |
|-----------|-------|---------|---------|----------|-----------------------------------------------------------|
| Hexane    |       |         |         |          | 2,5-Hexandion: 5 mg/g<br>Creatinine urine end of<br>shift |

| Component | Gibraltar | Latvia | Slovak Republic                                                                                                                                  | Luxembourg | Turkey |
|-----------|-----------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| Hexane    |           |        | 2,5-Hexanedione: 5<br>mg/L urine end of<br>exposure or work shift<br>4,5-Dihydroxy-2-hexano<br>ne: 5 mg/L urine end of<br>exposure or work shift |            |        |

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## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry

MDHS 99 Metals in air by ICP-AES

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                    | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Hexane<br>110-54-3 ( 75.00 ) |                              |                                 |                                | DNEL = 11mg/kg<br>bw/day          |

| Component                    | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Hexane<br>110-54-3 ( 75.00 ) |                                  |                                     |                                    | DNEL = 75mg/m <sup>3</sup>            |

## Predicted No Effect Concentration (PNEC)

No information available.

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material              | Breakthrough time                    | Glove thickness | EU standard | Glove comments (minimum requirement) |
|-----------------------------|--------------------------------------|-----------------|-------------|--------------------------------------|
| Nitrile rubber<br>Viton (R) | See manufacturers<br>recommendations | -               | EN 374      |                                      |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

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|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387 |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system.                                                                                                                                                                                                                             |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Colorless                |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | Highly flammable         | Estimated                                |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Immiscible               |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Hexane                                         | 4.11                     |                                          |
| <b>Vapor Pressure</b>                          | <=1100 hPa @ 50 °C       |                                          |
| <b>Density / Specific Gravity</b>              | 0.782 g/cm3              | @ 20 °C                                  |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### 9.2. Other information

|                                                                               |                                                       |
|-------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Molecular Formula</b>                                                      | C12 H27 Al                                            |
| <b>Molecular Weight</b>                                                       | 198.33                                                |
| <b>Explosive Properties</b>                                                   | Vapors may form explosive mixtures with air           |
| <b>Substances/mixtures which, in contact with water, emit flammable gases</b> | Emitted gas ignites spontaneously Gas(es) = Isobutane |

## SECTION 10: STABILITY AND REACTIVITY



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## 10.1. Reactivity

Yes

## 10.2. Chemical stability

Air sensitive. Moisture sensitive.

## 10.3. Possibility of hazardous reactions

Hazardous Polymerization  
Hazardous Reactions

No information available.  
None under normal processing. Reacts violently with water.

## 10.4. Conditions to avoid

Exposure to moist air or water. Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition.

## 10.5. Incompatible materials

Oxidizing agent.

## 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Metal oxides. Isobutane.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

#### Toxicology data for the components

| Component | LD50 Oral              | LD50 Dermal                  | LC50 Inhalation              |
|-----------|------------------------|------------------------------|------------------------------|
| Hexane    | LD50 = 25 g/kg ( Rat ) | LD50 = 3000 mg/kg ( Rabbit ) | LC50 = 48000 ppm ( Rat ) 4 h |

#### (b) skin corrosion/irritation;

Category 1 B

#### (c) serious eye damage/irritation;

Category 1

#### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

#### (e) germ cell mutagenicity;

No data available

#### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

#### (g) reproductive toxicity;

Category 2

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(h) STOT-single exposure; Category 3

Results / Target organs Central nervous system (CNS).

(i) STOT-repeated exposure; Category 2

Target Organs Central nervous system (CNS), Peripheral Nervous System (PNS).

(j) aspiration hazard; Category 1

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## 11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                               | Water Flea          | Freshwater Algae |
|-----------|---------------------------------------------------------------|---------------------|------------------|
| Hexane    | LC50: 2.1 - 2.98 mg/L, 96h flow-through (Pimephales promelas) | EC50: 3.87 mg/L/48h |                  |

### 12.2. Persistence and degradability

#### Persistence Degradation in sewage treatment plant

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
Immiscible with water, May persist.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Hexane    | 4.11    | No data available             |

### 12.4. Mobility in soil

Spillage unlikely to penetrate soil The product is insoluble and floats on water Is not likely mobile in the environment due its low water solubility.

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

ALFAA89057

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**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

14.1. UN number

UN3394

14.2. UN proper shipping name

ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE

Technical Shipping Name

Triisobutylaluminum, Hexane

14.3. Transport hazard class(es)

4.2

Subsidiary Hazard Class

4.3

14.4. Packing group

I

### ADR

14.1. UN number

UN3394

14.2. UN proper shipping name

ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE

Technical Shipping Name

Triisobutylaluminum, Hexane

14.3. Transport hazard class(es)

4.2

Subsidiary Hazard Class

4.3

14.4. Packing group

I

### IATA

14.1. UN number

UN3394

14.2. UN proper shipping name

ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE

Technical Shipping Name

Triisobutylaluminum, Hexane

14.3. Transport hazard class(es)

4.2

Subsidiary Hazard Class

4.3

ALFAA89057

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## 14.4. Packing group

I

## 14.5. Environmental hazards

Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

## 14.6. Special precautions for user

No special precautions required.

## 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                       | CAS No   | EINECS    | ELINCS    | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|---------------------------------|----------|-----------|-----------|-----|-------|------|----------|------|------|
| Hexane                          | 110-54-3 | 203-777-6 | 438-390-3 | -   | X     | X    | KE-18626 | X    | X    |
| Aluminum, tris(2-methylpropyl)- | 100-99-2 | 202-906-3 | -         | -   | X     | X    | KE-34851 | X    | X    |

| Component                       | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|---------------------------------|----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Hexane                          | 110-54-3 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Aluminum, tris(2-methylpropyl)- | 100-99-2 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component                       | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hexane                          | 110-54-3 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Aluminum, tris(2-methylpropyl)- | 100-99-2 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Seveso III Directive (2012/18/EC)

| Component                       | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Hexane                          | 110-54-3 | Not applicable                                                                            | Not applicable                                                                           |
| Aluminum, tris(2-methylpropyl)- | 100-99-2 | Not applicable                                                                            | Not applicable                                                                           |

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

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Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 2 (self classification)

| Component                       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|---------------------------------|---------------------------------------|-------------------------|
| Hexane                          | WGK2                                  |                         |
| Aluminum, tris(2-methylpropyl)- | nwg                                   |                         |

| Component | France - INRS (Tables of occupational diseases)            |
|-----------|------------------------------------------------------------|
| Hexane    | Tableaux des maladies professionnelles (TMP) - RG 59,RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                    | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Hexane<br>110-54-3 ( 75.00 ) | Prohibited and Restricted Substances                                                                           | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H250 - Catches fire spontaneously if exposed to air

H260 - In contact with water releases flammable gases which may ignite spontaneously

H304 - May be fatal if swallowed and enters airways

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

H336 - May cause drowsiness or dizziness

H361f - Suspected of damaging fertility

H373 - May cause damage to organs through prolonged or repeated exposure

H411 - Toxic to aquatic life with long lasting effects

EUH014 - Reacts violently with water

H225 - Highly flammable liquid and vapor

H315 - Causes skin irritation

## Legend

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**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

**Prepared By** Health, Safety and Environmental Department

**Creation Date** 19-Apr-2018

**Revision Date** 23-May-2024

**Revision Summary** New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**